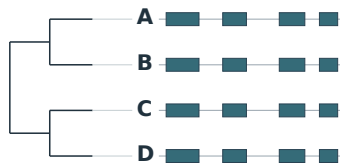


Genomic exon-span evolution on a supplied tree

Physical exon geometry, declared DNA material, and conditional phylogenetic summaries

A Orthologous loci



B Genomic correspondence

Whole-locus genomic alignment

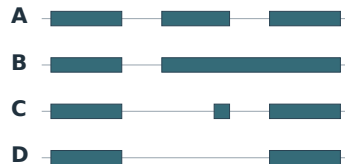


Shared genomic coordinates



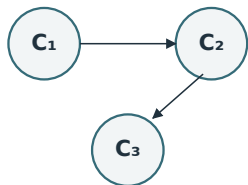
Sequence · strand · local order

C Physical exon spans



Physical intervals incl. UTR spans
One geometry per observation unit

D Admissible edits



Exon and DNA edits

E Likelihood inference



Exon-span CTMC

ML fitting

Tree pruning

One family rate

Supplied rooted species tree
Shared rate across edit kinds

F Branch summaries



Ancestral-state marginals

Branch: $P(\text{geometry differs})$
Joint modal parent $\rightarrow$ child